BOARD INVESTMENT REVIEW / 01

Invest for leadership. Preserve financial flexibility.

To: Board of Directors, Tiny-Rod Hit Inc.
From: Senior Finance Manager
Date: January 2025 | Decision horizon: upcoming fiscal year

Executive recommendation

Select Project B, the disruptive-technology R&D program, if only one venture is approved. Authorize its $40 million baseline investment only after technical, commercial and intellectual-property gates are passed. Earmark a further $5 million as a controlled contingency, not as automatic spending authority.

B combines the stronger illustrative expected value with a closer fit to the company's core capabilities. It uses $10 million less baseline capital than A and leverages the existing R&D team and customer market. Its 30% estimated failure probability is material: an unsuccessful program could lose the full $40 million with no significant revenue. Strong finances make that exposure affordable only if liquidity and stop/go discipline are protected. [1, 2]

Directional screen	Project A: new markets	Project B: core-market R&D
Baseline initial investment	$50m	$40m
Illustrative NPV at 9%	About +$8m	About +$25m, probability-adjusted
Illustrative IRR	About 12%	About 16% on expected cash flows
Critical trade-off	Diversification; thin value cushion	Leadership upside; binary failure risk

Estimates are scenario illustrations, not management forecasts. B's successful branch alone indicates roughly +$53m NPV and 23% IRR; the failed branch is approximately -$40m NPV. See pages 3–4 and 8.

Why this choice

- Value: B's expected NPV is roughly three times A's in the common ten-year screen; its expected-cash-flow IRR is about 7 percentage points above the 9% WACC, versus about 3 points for A.
- Execution fit: B builds on internal strengths. A adds a new product, unfamiliar geographies, local infrastructure and partnerships at the same time.
- Financial flexibility: a B-only $45m envelope, including reserve, leaves $55m of the assumed $100m pool unallocated. This is not automatically surplus cash.

If the Board requires both ventures

Allocate $55m to A and $45m to B: $50m and $40m of baseline funding, plus $5m of cash-held contingency for each. Deploy no more than $90m at initial commitment; retain the $10m reserve centrally. Proceed with both only after a corporate liquidity-floor test and separate project approval gates. Diversification complements B's technology exposure but does not eliminate downside.

Timing assumption: this is a January 2025 planning report. The $100m stated as available in May 2025 is treated as a forward funding assumption, to be verified before commitment; no January cash availability is inferred. [2]

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Strategic comparison

A stronger balance sheet expands the feasible set; it does not make every positive-growth proposal value creating.

Dimension	Project A	Project B
Investment thesis [1]	Launch a new product line in emerging markets; $50m upfront.	Develop a disruptive technology in the existing market; $40m initial investment.
Growth stated in reference [1]	20% annual revenue growth for the first five years; then 5%.	If successful, potential growth above 50% for the first three years, followed by rapid penetration.
Distinct strategic value	Geographic and customer-segment diversification; a platform for later products.	Defend market leadership; create differentiated IP and potentially redefine the industry.
Capability fit	Limited local experience; needs infrastructure and local partnerships.	Strong internal R&D team and existing core-market expertise.
Main exposure [1]	Geopolitics, regulation, currency and local competition.	R&D failure, obsolescence, established competitors and IP protection.
Portfolio implication	Reduces reliance on current markets, but adds unfamiliar execution risks.	Reinforces the core franchise, but increases dependence on its technology trajectory.

Value creation beyond the screen

A can create a repeatable distribution and local-partner platform. That option is valuable if the company can reuse infrastructure across future offerings, repatriate cash and sustain attractive unit economics. Those benefits are not quantified here. Country-specific capital traps, weak partner controls or price competition could instead turn revenue growth into low-return growth.

B can strengthen pricing power, customer retention and future product options. Existing customer access could shorten commercialization, but this is not assumed as a separate synergy. Cannibalization of current products, incremental support costs and competitor price responses must be deducted from the eventual business case; otherwise the technology's value will be overstated.

Why B ranks first, not why A should be abandoned

B's capability advantage and larger illustrative NPV compensate for its explicit failure risk in this screen. A's 20% growth is not evidence of lower overall risk: its country, regulatory and operational risks have no supplied probabilities. Keep A as a separately diligenced second investment rather than rejecting its long-term diversification value.

No supplied corporate strategy weights geographic diversification against core leadership. This report assumes both are legitimate objectives, with a near-term preference for exploiting proven capabilities. If the Board prioritizes reducing core-market concentration, A deserves a higher strategic weight, but that choice should be explicit rather than hidden in the valuation.

Source: project descriptions and risk factors from the reference file, page 1 [1]. Strategic interpretations are Finance judgments, not additional facts about the company.

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Valuation framework & assumptions

A directional investment screen, not a forecast or a completed discounted-cash-flow valuation.

Common measurement basis

Use nominal US-dollar, unlevered, after-tax incremental free cash flow (FCF), discounted at the supplied 9% WACC. FCF is cash after operating costs, taxes, sustaining investment and working-capital needs. Financing cash flows are excluded, avoiding a double count of debt benefits already reflected in WACC.

Time zero is the verified funding/authorization date, assumed no earlier than May 2025. Years 1–10 are the following ten 12-month periods, with year-end cash receipts. The full baseline outlay is charged at time zero for both projects. A starts generating revenue in Year 1; B has two pre-revenue development years and launches in Year 3. These timing assumptions require validation.

Model input	Project A	Project B: success branch
Initial outlay [1]	$50m	$40m
First revenue / scale [3]	Year 1: $25m revenue.	Year 3: $20m revenue; Years 1–2: zero.
Revenue path [1, 3]	20% increases in Years 2–5; 5% in Years 6–10.	60% increases in Years 4–6; 20% in Y7–Y10.
FCF conversion [3]	20% of revenue in each year.	25% of revenue in each revenue year.
Failure treatment [1, 3]	No failure probability supplied; use operating and cost stresses instead.	70% success / 30% failure. Failure yields zero operating FCF and no salvage.
Horizon / residual value [3]	Ten years; no terminal or salvage value.	Ten years; no terminal or salvage value.

The reference's growth rates are revenue rates, not cash-flow returns. First operating-year revenue anchors are invented for this screen. A's first five operating years are Years 1–5; B's first three post-launch year-on-year increases are Years 4–6. The 60% and later 20% rates for B are illustrative, not reference forecasts.

What the assumptions include—and do not prove

The initial budgets are assumed to cover all pre-revenue development and launch costs. The FCF conversion ratios are assumed net of ongoing reinvestment and working capital. If further R&D, infrastructure, tax or cash-conversion costs are needed, they must reduce value; none is financed implicitly with extra cash.

NPV equals discounted future FCF less the initial outlay. B's expected NPV weights the success and failure branches 70/30. A's case is not equivalently probability-weighted because its outcome probabilities are unknown. The comparison is therefore a screening judgment, not a fully risk-normalized ranking.

No perpetual value is attached to A's 5% revenue growth or B's disruptive potential. This avoids a terminal-value-driven decision but omits value after Year 10. It can disadvantage either project, especially A's long-lived market platform; a final appraisal must test finite useful lives, residual assets and sustainable terminal cash flows.

WACC and financial position: user brief [2]. All unsupplied model inputs: illustrative Finance assumptions [3]. A reproducible cash-flow bridge is on page 8.

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Directional returns & decision tests

Rounded indications only. Small numerical differences are not decision-grade.

Outcome / metric	Project A	Project B
Base NPV at 9%	About +$8m	Success: about +$53m Failure: about -$40m Expected: about +$25m
Base IRR	About 12%	Success: about 23% Expected-cash-flow IRR: about 16%
Value implication	Positive but only a modest cushion over cost of capital.	Larger expected cushion; does not guarantee a positive realized outcome.

The 16% figure is the IRR of B's probability-weighted cash-flow stream, not the probability-weighted average of realized IRRs. The zero-revenue failure branch has no economically meaningful IRR solution.

Sensitivity: one change at a time

Change from illustrative base	A NPV	B expected NPV
Base at 9%	~+$8m	~+$25m
All operating FCF 25% lower	~-$6m	~+$9m
All receipts delayed one year*	~-$2m	~+$8m
Initial outlay 20% higher	~-$2m	~+$17m
Discount rate increased to 12%	~$0m	~+$13m
B failure probability rises to 50%	Unchanged (~+$8m)	~+$6m

*The end of Year 10 remains fixed, so delayed receipts beyond that date are excluded. This stress captures the lost timing window as well as discounting. A two-year B delay makes its expected NPV roughly -$6m. All stresses are illustrative and not forecasts.

Practical decision thresholds

- A: roughly a 14% reduction in the present value of operating FCF consumes its entire base NPV. Its indicative IRR is close to a 12% stress hurdle; FX, pricing or setup costs could readily erase the margin.
- B: with the modeled successful cash flows, a success probability near 43% is needed merely to break even. About 52% is needed to match A's base NPV, versus the supplied 70% assumption. These are thresholds, not revised technical probabilities.
- B: if successful-case cash flows are 25% lower, expected NPV remains positive but is only about equal to A's base value. A simultaneous delay, weaker margins and lower success probability could make B unattractive.

Recommendation discipline

Choose B on expected enterprise value, capability fit and smaller baseline cash exposure—not on the headline 50%+ revenue growth or IRR alone. Require evidence supporting the revenue scale, conversion margin, two-year development timetable and 70% success assumption. If diligence lowers B's success probability to roughly one-half while A's case is validated, prefer A or defer commitment.

A 12% discount rate is an analytical stress, not an independently estimated project WACC. Probability weighting captures modeled R&D failure; it does not capture every technology or execution risk.

BOARD INVESTMENT REVIEW / 05

Project B: top three risks

Risk priority reflects potential cash loss and strategic damage. Controls are proposed, not existing company practices.

1 Technical failure or technology obsolescence

Exposure: the supplied 30% failure chance implies no significant revenue, and up to $40m of baseline capital could be lost. A breakthrough arriving too late can also become commercially obsolete even if the laboratory work succeeds. [1]

Mitigation: the CTO should commission an independent architecture and feasibility review before authorization; establish measurable prototype, performance, reliability and unit-cost milestones. Benchmark alternatives quarterly and protect a small, separately approved redesign option. Release cancellable work packages only against evidence where contracts and the development architecture permit.

Trigger and contingency: a failed critical benchmark, or two consecutive milestone misses, pauses new commitments. The CTO and CFO return to the Investment Committee within 30 days with a re-engineered, licensed or partnered alternative and a refreshed NPV. Escalate if success confidence approaches the 43% initial-investment break-even benchmark. Continue only if remaining avoidable spending has positive forward NPV; otherwise stop, salvage transferable IP and redeploy staff. No salvage proceeds are assumed.

2 Commercial adoption, competition and IP leakage

Exposure: slow adoption, incumbent price responses, cannibalization or weak IP protection can turn technical success into negative incremental cash flow. Cash projections depend on both revenue scale and sustained conversion margins. [1, 3]

Mitigation: the commercial lead should secure paid design-partner pilots, validate willingness to pay and require a bottom-up net revenue case after cannibalization. Legal should complete freedom-to-operate review, file defensible patents where appropriate, use trade-secret controls and audit partner access to source code and know-how.

Trigger and contingency: before scaling, require at least three paid pilots and documented margin evidence consistent with the modeled 25% FCF conversion. If pilot economics miss plan by more than 20%, reprice or narrow the use case and seek an OEM/licensing channel. Unresolved blocking IP or negative forward NPV halts launch; pursue cross-licensing only within an approved settlement cap.

3 Cost escalation, delivery delays and cash strain

Exposure: scarce talent, integration problems or underestimated commercialization costs can push spending beyond $40m and postpone receipts. One year of delay cuts expected NPV to about +$8m; two years makes it about -$6m in the fixed-horizon screen.

Mitigation: the CFO and program lead should set a cost-to-complete baseline, retain key staff, maintain backup suppliers and report monthly cash burn, schedule variance and customer-readiness status. Keep the $5m contingency centrally controlled; contractual cancellation rights matter more than accounting budget labels.

Trigger and contingency: a forecast overrun above 10% of the $40m budget, or a delay exceeding one quarter, requires an immediate reforecast and freeze on discretionary scope. De-scope to a minimum viable product or phase the launch. Seek a co-development partner if that preserves IP and value; stop rather than fund a negative-NPV rescue. Any reserve draw or further capital requires CFO review and Board approval.

Thresholds, owners and response times above are proposed governance assumptions. Stage gates cannot recover sunk spending; modeled failure conservatively assumes the full baseline outlay is lost.

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If both ventures must proceed

Fund distinct strategic engines, but keep project approval and corporate liquidity tests separate.

Allocate the full $100m as controlled envelopes

Use of cash	A: new markets	B: technology	Total
Baseline investment	$50m	$40m	$90m
Earmarked contingency, held centrally	$5m	$5m	$10m
Maximum envelope	$55m (55%)	$45m (45%)	$100m

For this alternative, assume both ventures are viable. The allocation meets both baseline funding requirements; a smaller investment is not assumed to buy a proportionate share of the same revenue. A receives more dollars because it requires more upfront capital, not because it ranks above B. The reserves are optional risk capacity, not planned capex; the Board can reassign them if the relative economics change.

Commit in sequence; do not dilute the required funding

Before release, complete B's technical/IP review and secure pilot-customer commitments; complete A's country, regulatory and partner diligence. A's $50m is assumed required upfront at its launch decision; B's full $40m is assumed required at authorization. If B's work packages are genuinely divisible and cancellable, milestone-based draws can reduce early exposure. Do not rely on that possibility to fund both, and do not treat ring-fenced commitments as free cash.

Base funding of $90m leaves only $10m gross cash in the assumed pool. Treasury must first validate a minimum liquidity floor covering operating needs, seasonal working capital, debt service and a downside cash forecast. That floor cannot be quantified from the brief. The $10m project reserve is not a substitute for operating liquidity, and it cannot be spent twice.

If the floor exceeds $10m, defer A's commitment until operating cash generation replenishes funds, or secure a committed undrawn credit facility before authorizing both. Use debt only after testing covenant headroom, downside debt service and funding costs; a healthy debt-to-equity ratio is not evidence of unlimited capacity. Do not approve simultaneous full commitments without that solution.

Portfolio value and boundary conditions

- Long-term value: A builds a new geographic growth platform while B protects and renews the core franchise. The two programs address different sources of future relevance.
- Diversification: country/customer exposure differs from R&D failure exposure, but both remain vulnerable to weak demand, competition, talent shortages and execution overload. No numerical correlation benefit is assumed.
- Capital discipline: simple addition gives about +$33m of illustrative combined expected NPV on $90m baseline investment, before shared constraints, cannibalization or synergies. Diversification does not itself increase that arithmetic NPV.

Appoint separate sponsors and reserve shared engineering, commercial and compliance capacity explicitly. Protect B's critical milestones from A's setup demands. Unused contingency stays cash; if all $10m were instead spent at time zero with no extra benefit, combined NPV would fall from about +$33m to about +$23m.

Under a B-only decision, the proposed $45m envelope leaves $55m unallocated in the $100m pool. Under the both-project decision, no additional unallocated cash remains beyond the $10m earmarked reserves.

BOARD INVESTMENT REVIEW / 07

Approval gates & Board resolutions

Use strong financial health to buy disciplined options, not to weaken investment hurdles.

Gate / owner	Required evidence	Board response
Before May funding CFO / Treasury	Confirm actual cash, restrictions, debt covenants, liquidity floor and downside headroom.	No binding commitment before cash and funding capacity are verified.
B authorization CTO / Commercial / Legal	Independent technical review; pilot-customer commitments; IP clearance; integrated cost-to-complete and staffing plan.	Approve $40m only if updated expected NPV is positive and strategically compelling.
A authorization, if both International lead / Legal / CFO	Country-level economics, licenses, cash repatriation, local competition, partner diligence and infrastructure budget.	Approve the full $50m only if it clears the hurdle and does not compromise corporate liquidity.
Quarterly review Investment Committee	Incremental cash forecast; technical/customer progress; forecast-at-completion; liquidity and cross-project resource use.	Continue, re-scope, partner, defer or stop using forward cash flows, not sunk-cost recovery.

Information required before final approval

Replace assumed starting revenues with a customer/volume/price build; substantiate margins, tax and FX exposures; identify incremental capex, maintenance, working capital and any future funding rounds. For B, validate success probability, development duration, cannibalization and IP economics. For A, build country-weighted scenarios, local partner terms, profit repatriation and currency sensitivities.

Prepare integrated downside cases, not only one-variable stresses: lower revenue plus delay plus higher costs, with corporate cash and debt capacity tracked monthly. Validate useful lives and post-Year-10 cash flows so that a long-lived infrastructure platform and a potentially short-lived technology are compared fairly. Revisit project-specific discount rates only with a consistent treatment of cash-flow risk.

Resolutions proposed for the minutes

- Preferred path: endorse B as the single-project priority, conditional on the gates above; authorize a $40m baseline and a separate $5m contingency envelope after diligence and Treasury sign-off.
- Alternative if both are required: reserve $55m for A and $45m for B, inclusive of $10m central contingencies. Approve each baseline independently and sequence commitments if the liquidity-floor test is not met.
- Control: require monthly program dashboards, quarterly investment reviews, and Board approval of reserve use or any increase in the total envelope. Suspend discretionary spending when the page 5 triggers are breached.

Final judgment

B offers the better current combination of illustrative expected value, strategic renewal and execution fit. A remains a credible second engine of growth and diversification, but its narrower value cushion requires stronger proof of cash conversion and local execution. A healthy balance sheet supports either choice—and potentially both—only when capital is released against evidence and liquidity is preserved.

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Appendix: model bridge & source notes

Audit trail for the rounded estimates. All amounts below are nominal USD millions.

Illustrative operating cash flows

Year after funding	A FCF	B success FCF	B expected FCF
0: initial investment	(50)	(40)	(40)
1	5.0	0.0	0.0
2	6.0	0.0	0.0
3	7.2	5.0	3.5
4	8.6	8.0	5.6
5	10.4	12.8	9.0
6	10.9	20.5	14.3
7	11.4	24.6	17.2
8	12.0	29.5	20.6
9	12.6	35.4	24.8
10	13.2	42.5	29.7

Displayed cash flows are rounded for explanation; do not interpret them as precise forecasts. Parentheses denote outflows. B expected operating FCF is 70% of its success branch; the 30% failed branch contributes zero. B's initial $40m outlay is certain in every branch.

How the directional numbers connect

A's revenue starts at $25m and its 20% FCF conversion produces $5m in Year 1. Growth takes revenue to roughly $52m in Year 5 and $66m in Year 10. Its operating FCF has a present value around $58m at 9%, leaving about +$8m NPV after the $50m outlay.

B's successful revenue starts at $20m in Year 3, reaches about $82m in Year 6 and $170m in Year 10. Its 25% FCF conversion yields operating cash-flow present value around $93m. Probability weighting reduces that to about $65m; subtracting $40m gives expected NPV of about +$25m. Equivalently: 70% × success NPV + 30% × failure NPV.

Sources, provenance and limitations

[1] Tiny Rod Hit Inc. Reference, supplied one-page PDF, page 1, Project A and Project B descriptions. Source for project budgets, stated revenue growth, B's 30% failure probability, risk factors and capability/strategic descriptions.

[2] User-provided Board assignment. Source for January 2025 planning context, $100m available as of May 2025, consistent profitability, strong balance sheet, healthy debt-to-equity ratio and 9% WACC. No actual debt amounts, cash forecast or covenant headroom were provided.

[3] Finance illustrative assumptions for this report: starting revenue, launch timing, conversion margins, ten-year horizon, zero residual value, B's 60%/20% growth choices, stress cases, reserves and governance thresholds. These are not supplied forecasts, empirical probabilities or approved budgets. No external market data was used.

The report is a high-level capital-allocation recommendation, not a valuation opinion or assurance of future returns. All approvals remain subject to replacement of illustrative inputs with validated incremental cash-flow forecasts.